
CS 685: Natural Language Processing

Course Code	CS 685
Course Name	Natural Language Processing
L-T-P-C	3-0-0-3
Prerequisites	Introductory course in machine-learning/AI; Programming in Python
Intended for	BTech 3rd/4th year; MTech; MSc; Ph.D.
Approval	54th BoA (May 2024)

Course Contents

- Introduction to NLP and FSTs (6 Hours)
 - History of NLP and its Evolution
 - Challenges and Applications of NLP
 - Text Preprocessing: Tokenization, Stop words removal, Stemming, Lemmatization, and Normalization
 - Parsing Algorithms
 - Regular Languages and Finite State Transducers
- Language Modeling, Word Embeddings and HMM (12 Hours)
 - Language modeling with Markov chains
 - Introduction to probability and information theory in NLP
 - Concept of Perplexity
 - Text Classification: Classification Layers, Low-Rank Adaptation (LoRA)
 - Word2Vec, CBOW, Skip-gram, and FastText
 - GloVe: Global Vectors for Word Representation
 - HMM and Viterbi
- Text Generation (10 Hours)
 - Using pre-trained word embeddings for specific applications
 - Sentence Embeddings
 - Introduction to text generation
 - Sequence-to-Sequence Models, Attention Mechanism and Transformers
 - Recurrent Neural Networks (RNNs) for text generation Large Language Models
 - Language generation with GPT-3 and similar models
 - Ethical considerations, future trends and research directions in NLP
- Named Entity Recognition (NER) and Part-of-Speech Tagging (POS) (9 Hours)
 - Introduction to NER
 - Sequence labeling for NER
 - CRF-based NER models
 - Deep learning approaches to NER
 - Introduction to POS tagging
 - Rule-based POS tagging
 - Hidden Markov Models for POS tagging
 - Neural network-based POS tagging
- Text Post Processing and Sentiment Analysis (8 Hours)

- Coreference resolution and coherence
- Adaptive Text Post Processing using plugin classifiers
- Understanding sentiment analysis and Lexicon-based approaches
- Machine learning-based sentiment analysis

Laboratory Modules

In addition to the lecture modules, this course will include a take home assignment where students will apply their knowledge to develop and fine-tune LLM applications using Python and platforms such as Hugging Face Transformers.

Text Books

1. Yoav Goldberg, **Neural Network Methods for Natural Language Processing**, Morgan and Claypool, 2017.
2. Dan Jurafsky and James, **Martin Speech and Language Processing**, 3rd Edition.

References

1. Allen, James, **Natural Language Understanding**, 2nd Edition, Benjamin/Cumming, 1995.
2. Charniack, Eugene, **Statistical Language Learning**, MIT Press, 1993.
3. Jurafsky, Dan and Martin, James, **Speech and Language Processing**, 2nd Edition, Prentice Hall, 2008.
4. Manning, Christopher and Heinrich, Schutze, **Foundations of Statistical Natural Language Processing**, MIT Press, 1999.